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Internship Domain: Data Science
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Objective

The primary objective of this project is to build a comprehensive data cleaning, preprocessing, and feature engineering pipeline on a public railroad highway-rail crossing accident dataset using Python. The goal is to clean noisy records, handle missing values, mitigate extreme outliers, engineer new domain-specific features, and format categorical/numerical variables to prepare the dataset for predictive machine learning models (e.g., predicting accident severity, injury outcomes, or damage costs).

Problem Statement

Raw highway-rail crossing collision records maintained by transportation authorities often suffer from severe data quality issues. These include missing environmental/speed records, duplicate incident reports, inconsistent formatting, extreme outliers in damage costs and vehicle speeds, and unencoded categorical fields (such as weather conditions, state names, and equipment types).

If unaddressed, these flaws lead to inaccurate statistical analyses, skewed model parameters, and runtime failures during model training. Building an automated cleaning and feature engineering pipeline resolves these defects and unlocks key risk insights.

Technologies Used:

Python, Pandas, NumPy, Matplotlib & Seaborn, Google Colab, Scikit-learn.

Dataset Description:

It consists of exactly 239485 records and 149 features.

The dataset contains historical railroad highway-rail crossing collision reports. It spans multi-dimensional features across operational, driver, environmental, and outcome categories:

- **Temporal & Location Features:** `date`, `time`, `incident_datetime`, `state_name`.
- **Driver & Vehicle Attributes:** `user_age`, `user_gender`, `highway_user`, `estimated_vehicle_speed`, `number_vehicle_occupants`.
- **Train & Railway Attributes:** `train_speed`, `railroad_type`, `equipment_type`, `number_of_cars`, `number_of_locomotive_units`.
- **Environmental Attributes:** `weather_condition`, `temperature`.
- **Outcome & Impact Metrics:** `vehicle_damage_cost`, casualty counts (users/employees/passengers killed or injured)

Dataset Source:

<https://www.kaggle.com/datasets/yogidsba/us-highway-railgrade-crossing-accident>

Preprocessing (Step-by-Step Pipeline)

1. **Duplicate Handling:** Identified and removed identical duplicate records using `df.drop_duplicates()` to avoid duplicate counting of incidents.
2. **Column Standardization & String Cleaning:** Converted column names to lower_snake_case format and replaced blank whitespace strings with proper `NaN` missing values.
3. **Data Type Correction:** Converted `date` into pandas `datetime` format and explicitly cast numerical metrics (`user_age`, `train_speed`, `vehicle_damage_cost`, etc.) into float/integer types.
4. **Missing Value Imputation:** Applied median imputation for numerical features (resistant to extreme values) and substituted missing categorical entries with `"Unknown"`.
5. **Outlier Detection & Capping:** Identified extreme outliers in speed, age, and cost attributes using Boxplots and capped them using the Interquartile Range (IQR) method ($\$Q1 - 1.5 \times \text{IQR}$ to $\$Q3 + 1.5 \times \text{IQR}$).
6. **Feature Engineering:** Derived **8 new domain features** to enhance predictive performance:
 - `incident_datetime`: Combined `date` and `time`.
 - `is_night`: Binary indicator (1 if hour ≤ 6 or ≥ 18 , else 0).
 - `is_weekend`: Binary flag (1 for Saturday/Sunday).
 - `speed_difference`: `estimated_vehicle_speed` - `train_speed`.
 - `total_injuries`: Sum of user, employee, and passenger injury counts.
 - `total_fatalities`: Sum of user, employee, and passenger fatality counts.
 - `damage_per_occupant`: `vehicle_damage_cost` divided by occupant count.
 - `accident_severity`: Weighted severity score ($(\text{total_injuries} + 3 \times \text{total_fatalities})$).
7. **Encoding & Standardization:** Encoded categorical features (`state_name`, `weather_condition`, etc.) using `LabelEncoder` and standardized continuous numerical attributes via `StandardScaler`.

Conclusion:

The 42-cell preprocessing pipeline transformed a raw, uncleaned railroad crossing dataset into an analysis-ready dataset. Through deduplication, robust median imputation, IQR outlier capping, label encoding, and standardization, all 0 missing values remain.

Additionally, creating 8 meaningful domain features significantly improves the dataset's information density, making it fully prepared for downstream classification and regression models.

GitHub Repository Link:

<https://github.com/abdulwahab014-CS/USRAODACCIDENT>